Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 23-03-2026

Relational Inventory Control & Stock Tracking System

SDG: SDG No – SDG 9 – Industry, Innovation and Infrastructure

Presented By:

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Courses Handling Faculty:

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INTRODUCTION

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- Many businesses manage inventory manually, which leads to errors, inaccurate stock records, and difficulty in tracking product movement.
- There is a need for a relational database system to efficiently control inventory, track stock in real-time, and reduce overstocking or understocking problems.
- This project is important because it helps businesses accurately manage inventory, reduce errors, and track stock efficiently in real time.
- It is widely used in supermarkets, warehouses, pharmacies, and e-commerce companies to manage and track stock efficiently.
- It helps businesses prevent losses by avoiding overstocking and stock shortages in real-time operations.
- The Relational Inventory Control & Stock Tracking System is a database-based system used to manage and monitor inventory efficiently.
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PROBLEM STATEMENT

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- In many businesses, inventory management is handled manually or using basic spreadsheets.
- This leads to errors in stock calculation and delays in updating records.
- It becomes difficult to track product movement and monitor real-time stock levels.
- Overstocking and understocking problems often occur due to lack of proper monitoring.
- Manual systems also make reporting and data retrieval time-consuming.
- There is a need for an automated and reliable system to manage inventory efficiently.
- A relational database-based solution can improve accuracy and organization.
- The proposed system aims to provide real-time stock tracking and better inventory control.

OBJECTIVE(S)

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1. To design and develop a relational inventory control system for real-time stock tracking with high accuracy.
2. To implement automated reorder alerts and reporting to improve inventory efficiency and reduce stock errors.

PROPOSED SYSTEM

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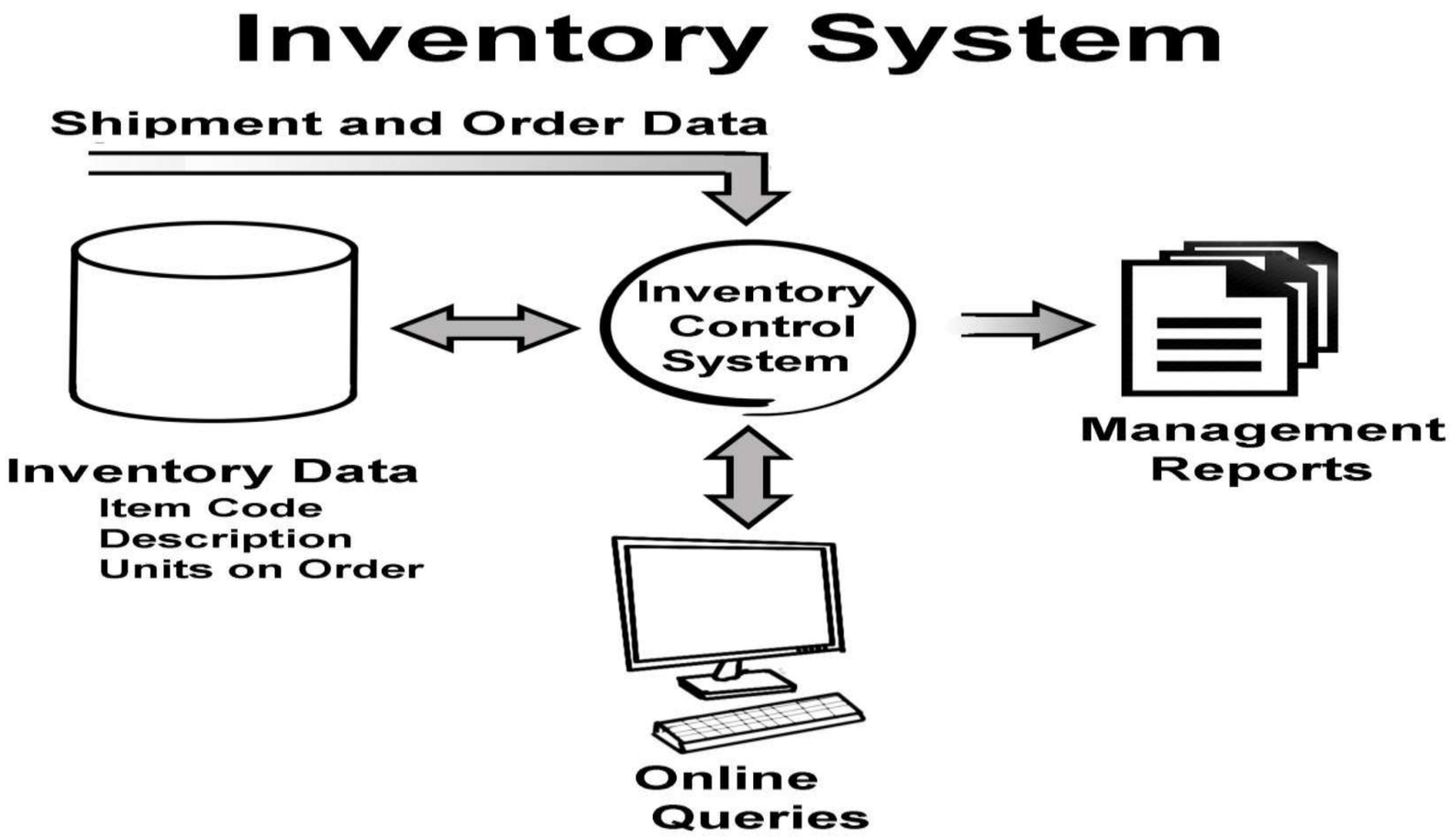
- A single system to manage products, stock, suppliers, and orders in one place.
- Real-time stock updates so you always know current inventory levels.
- Automatic alerts when stock is low to avoid shortages or excess stock.
- Keeps a record of all transactions for better accuracy and tracking.
- Connects purchasing, sales, and inventory to avoid confusion and errors.
- Provides quick reports to help in better decision-making.
- Reduces manual work and mistakes, saving time and effort.
- Secure and scalable system that can support multiple locations and users.

PROPOSED SYSTEM ARCHITECTURE



WORKFLOW

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TECHNOLOGIES USED

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Frontend:
1.HTML5
2.CSS3
3.Javascript

Backend:
1.Java Spring Boot

Database:
MySQL

Server:
XAMPP (Apache Server + MySQL)

Tools:
1. Web Browser (Chrome / Edge)
2. VS Code

MODULE DESCRIPTION

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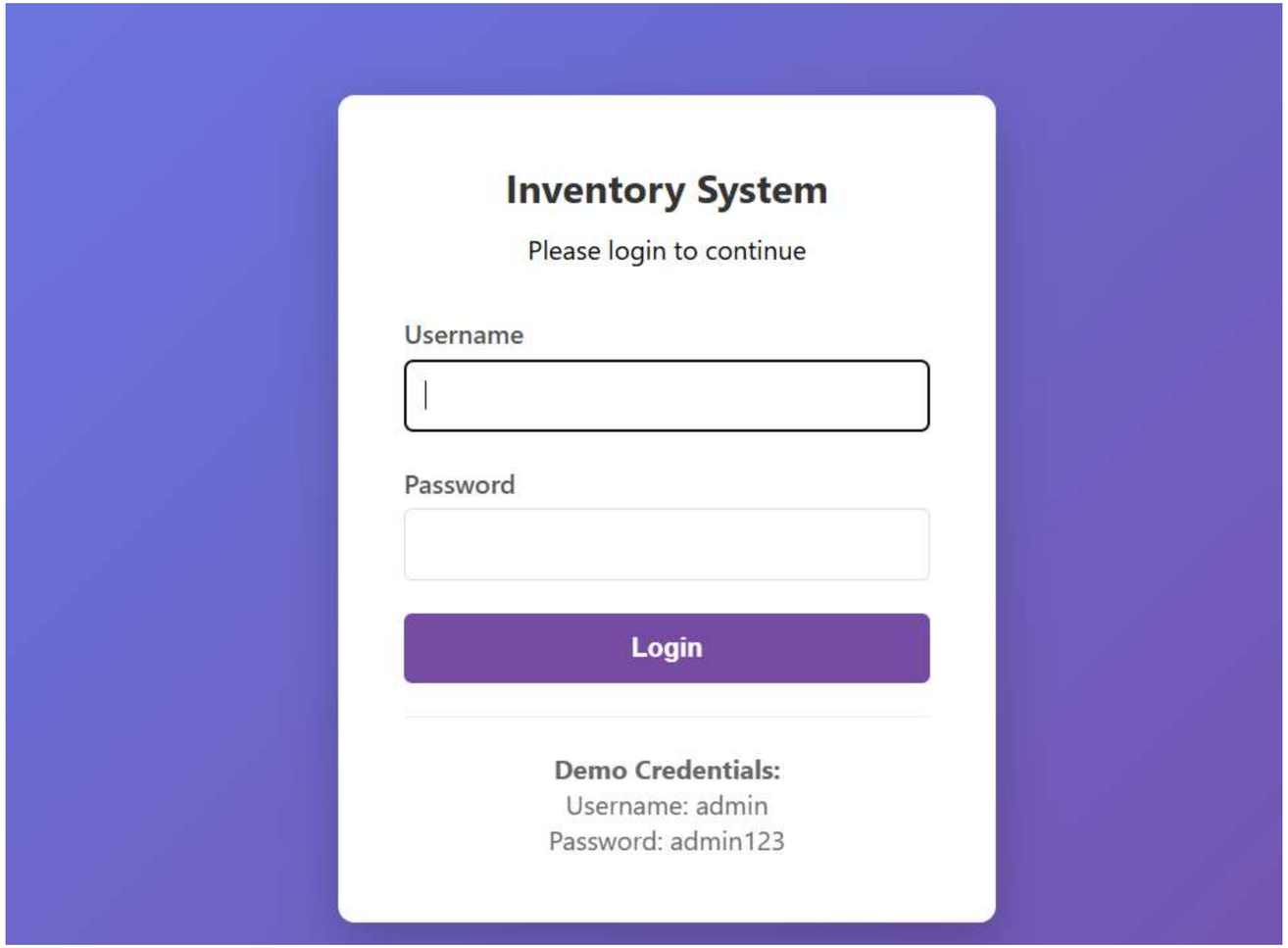
Module 1: User Management
Handles user registration, login, and profile management with secure authentication.
Module 2: Core Processing
Implements the main business logic and processes user inputs efficiently.
Module 3: Database Management
Manages data storage, retrieval, and ensures data integrity.
Module 4: Reporting & Analytics
Generates reports and visual insights for better decision-making.

IMPLEMENTATION SCREENSHOTS



IMPLEMENTATION SCREENSHOTS

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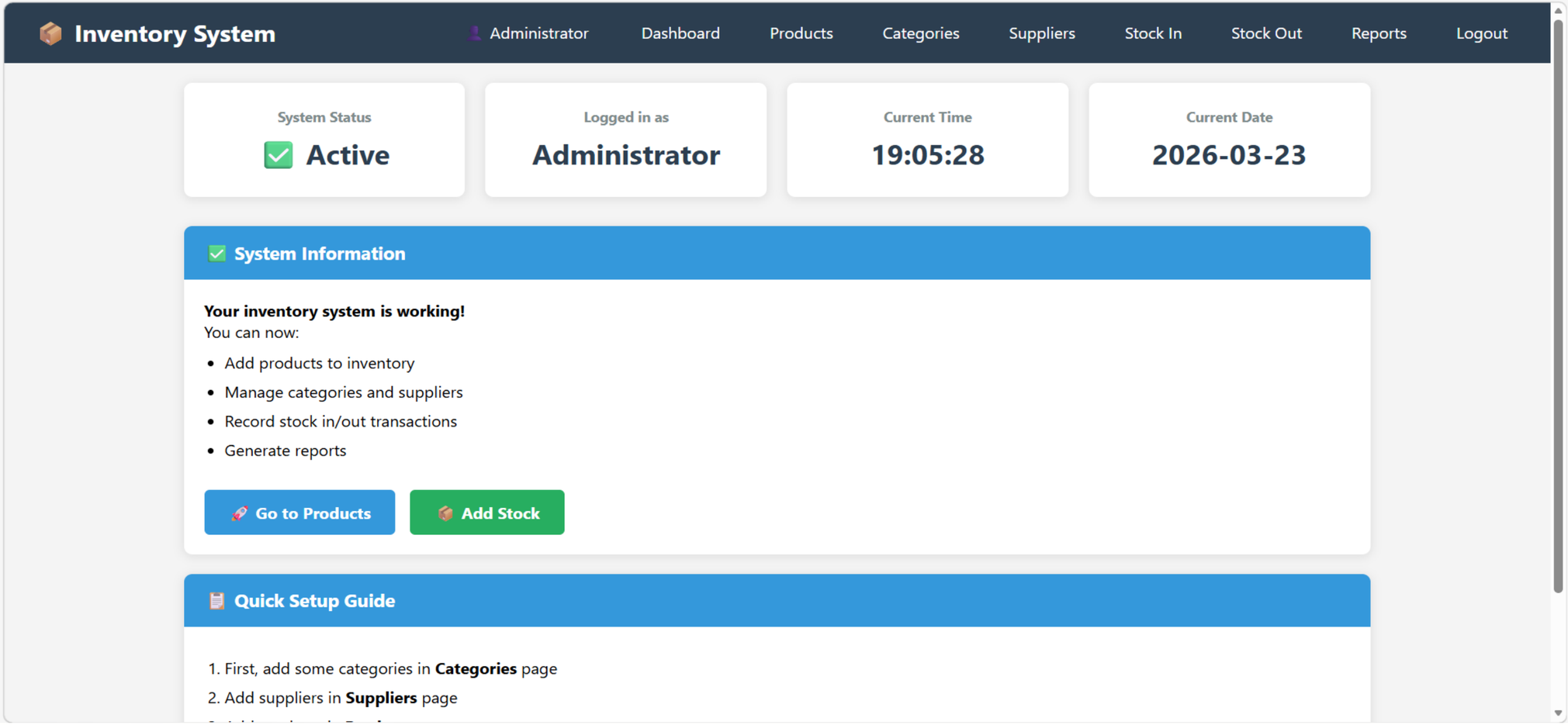

IMPLEMENTATION SCREENSHOTS

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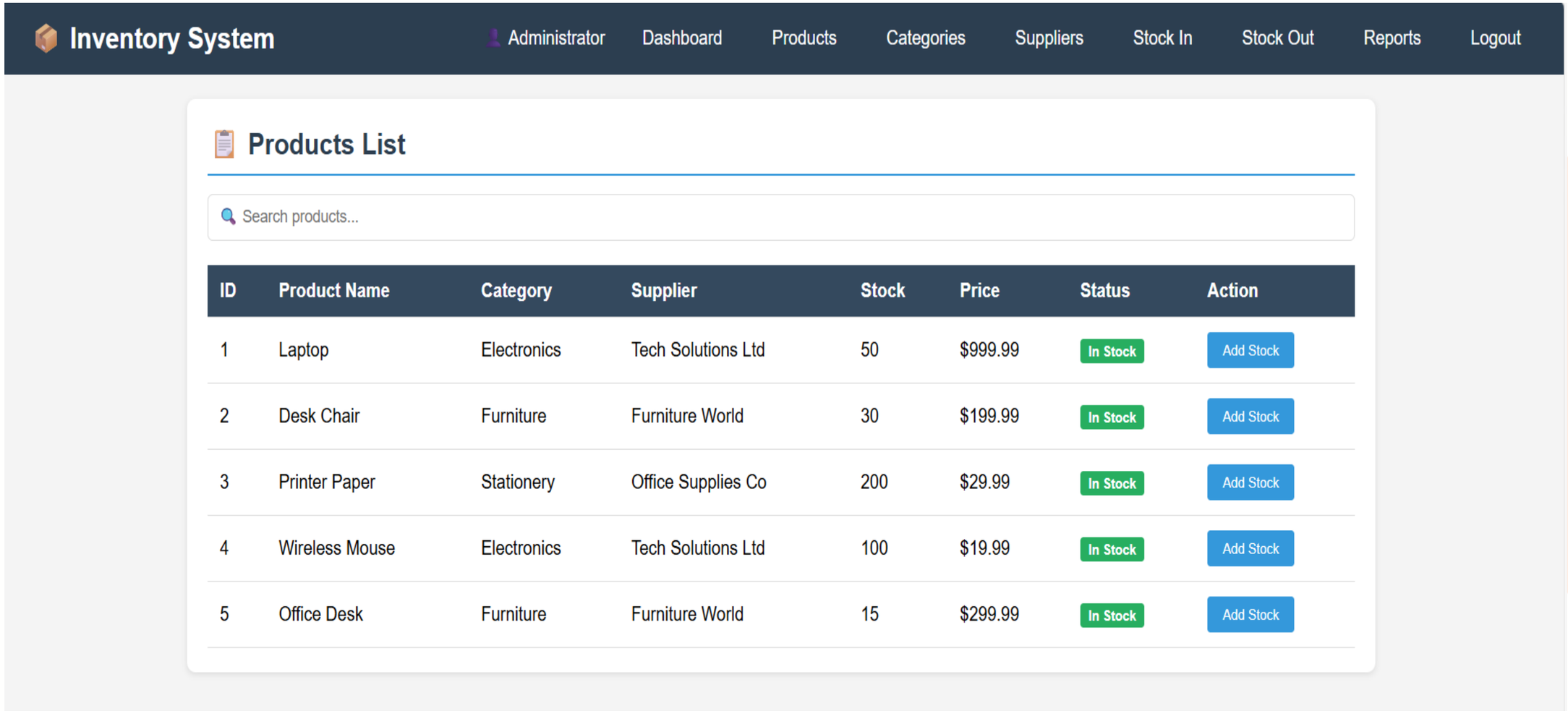

IMPLEMENTATION SCREENSHOTS

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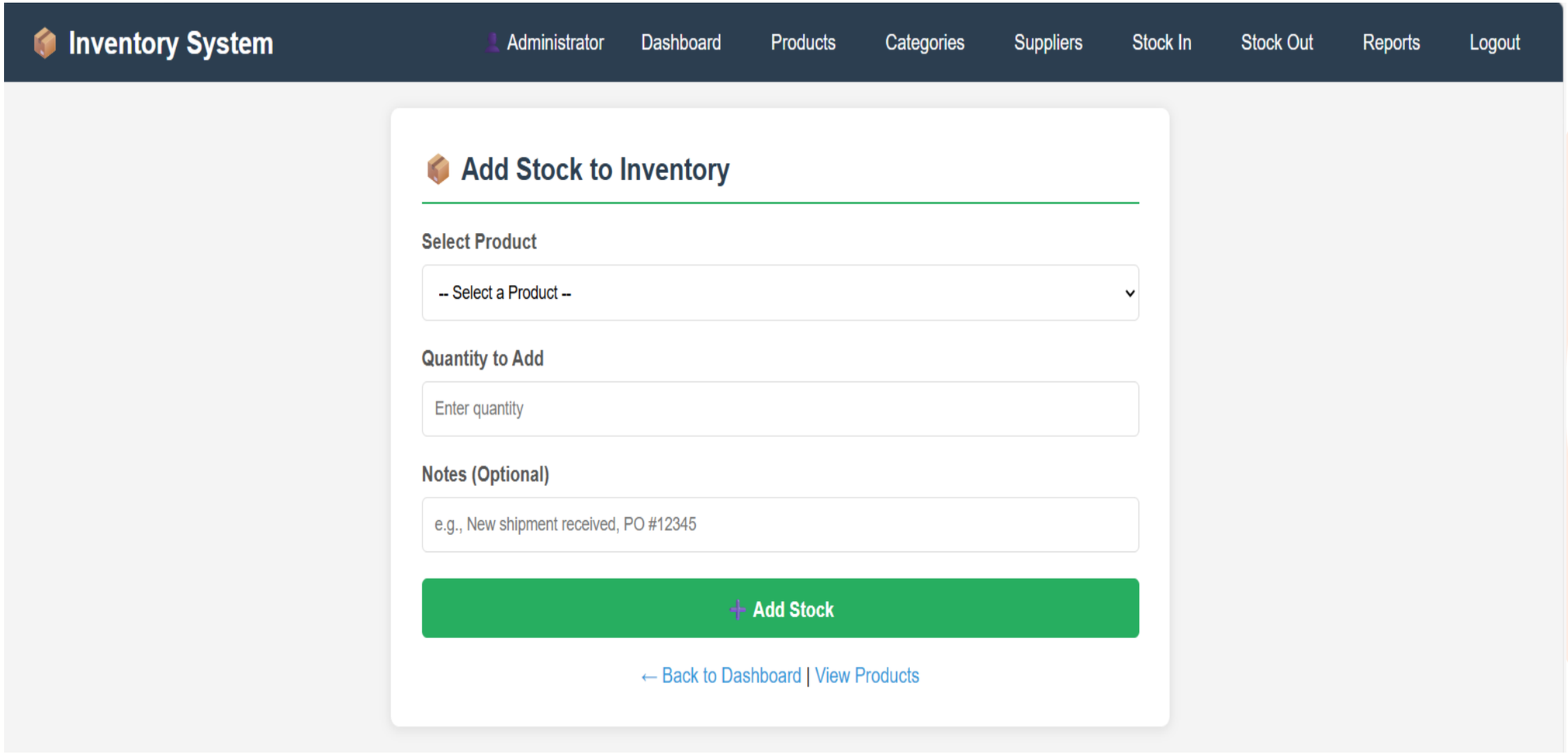

ADVANTAGES & LIMITATIONS

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ADVANTAGES:

1. Reduces human errors in stock management
2. Real-time tracking of inventory levels
3. Saves time by automating tasks

LIMITATIONS

1. Initial setup cost can be high
2. Requires technical knowledge to manage
3. Dependent on internet/system availability
4. Data loss risk if backup is not maintained
5. Training required for users
6. May need regular maintenance and updates

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Thank you